Analytic and Native-Space Route

This note records a speculative but concrete route suggested by the GGRT stability paper and by the analyticity of the torsion function inside the domain.

The guiding idea is:

> The STFT proof works because the optimizer lives in a native global analytic space, the Bargmann–Fock space. For torsion, one should look for the closest native replacement: harmonic functions, modified complex gradients in dimension two, and boundary graph spaces after the BDV selection step.

The goal is not to replace the BDV route immediately, but to find a setting in which a GGRT-style second variation can be computed cleanly.

1 The Harmonic Replacement of the Torsion Function

Let

$$-\Delta u = 1 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

Define

$$h(x) := u(x) + \frac{|x|^2}{2N}.$$

Then

$$\Delta h = 0 \quad \text{in } \Omega.$$

For a ball B_R centered at the origin,

$$u_B(x) = \frac{R^2 - |x|^2}{2N}, \qquad h_B(x) = \frac{R^2}{2N}.$$

Thus the ball is characterized by the fact that the harmonic function h is constant on the domain. On the boundary of a general domain,

$$h|_{\partial\Omega} = \frac{|x|^2}{2N}.$$

If $\partial\Omega$ is a small graph over ∂B_R,

$$x = (R + \varphi(\theta))\theta,$$

then

$$h|_{\partial\Omega} = \frac{R^2}{2N} + \frac{R}{N}\varphi(\theta) + O(\varphi^2).$$

After pulling back to B_R, the first-order part of $h - h_B$ is just the harmonic extension of $(R/N)\varphi$. This is exactly why the BDV second variation sees the $H^{1/2}$-norm of the boundary graph.

So the natural "native space" for the torsion problem is not an entire Fock space, but the harmonic Hardy/Dirichlet space on the ball, after the selected domain has been reduced to a graph.

2 Two-Dimensional Holomorphic Formulation

In dimension $N = 2$, there is a sharper analytic object. With

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y),$$

the equation $\Delta u = -1$ gives

$$\partial_{\bar{z}}\partial_z u = -\frac{1}{4}.$$

Therefore

$$F_\Omega(z) := \partial_z u(z) + \frac{\bar{z}}{4}$$

is holomorphic in Ω.

For the disk centered at the origin,

$$u_B(z) = \frac{R^2 - |z|^2}{4}, \qquad \partial_z u_B = -\frac{\bar{z}}{4},$$

so

$$F_B \equiv 0.$$

Thus F_Ω measures deviation from the disk in a genuinely holomorphic space. If Ω is a selected smooth near-disk domain, one can pull F_Ω back to the unit disk by a conformal map or by the normal graph. This suggests a possible Bergman/Hardy-space analogue of the GGRT Fock-space argument.

The boundary condition $u = 0$ implies that ∇u is normal to $\partial\Omega$. If the Serrin condition is nearly satisfied, then $|\nabla u|$ is nearly constant on $\partial\Omega$. In complex form, this places an approximate boundary constraint on F_Ω. This may allow one to turn the weighted Holder deficit D_H into a boundary norm for a holomorphic function.

3 Candidate GGRT-Style Functional for Torsion

For a fixed volume level s, define

$$\mathcal{K}_s(\Omega) := \int_{\{u_\Omega > u_\Omega^*(s)\}} u_\Omega \, dx.$$

The ball maximizes $\mathcal{K}_s$ among domains with fixed volume, at least morally by the same rearrangement/level-set proof. A GGRT-style variational route would try to prove, for nearly spherical sets,

$$\mathcal{K}_s(B) - \mathcal{K}_s(\Omega) \gtrsim_s \|\varphi\|^2_{H^{1/2}(\partial B)}$$

under the usual volume and translation orthogonality conditions.

This is a level-set-sensitive analogue of the BDV expansion for the full torsion energy

$$E(\Omega) - E(B) \gtrsim \|\varphi\|^2_{H^{1/2}}.$$

The advantage is that $\mathcal{K}_s$ interacts directly with the Nicola–Tilli convexity proof and with the matched superlevel set $\{u > u^*(s)\}$. The obstacle is that differentiating $\mathcal{K}_s$ moves both the domain and the internal level surface.

4 How to Compute the Second Variation

Work in the already-selected near-ball regime:

$$\partial\Omega_\varepsilon = \{(1+\varepsilon\varphi(\theta))\theta : \theta \in \partial B\},$$

with

$$\int_{\partial B} \varphi = 0, \qquad \int_{\partial B} \theta_i \varphi(\theta)\, d\theta = 0.$$

Let

$$u_\varepsilon = u_{\Omega_\varepsilon}, \qquad h_\varepsilon = u_\varepsilon + \frac{|x|^2}{2N}.$$

Pull h_ε back to B. To first order,

$$h_\varepsilon - h_B \approx \varepsilon H(\varphi),$$

where $H(\varphi)$ is the harmonic extension of a multiple of φ. Expanding in spherical harmonics,

$$\varphi = \sum_{k\geq 2} \varphi_k,$$

one expects a diagonal second variation

$$\mathcal{K}_s(B) - \mathcal{K}_s(\Omega_\varepsilon) = \varepsilon^2 \sum_{k\geq 2} a_k(s)\|\varphi_k\|^2_{L^2(\partial B)} + o(\varepsilon^2).$$

The key task is to prove a spectral gap

$$a_k(s) \geq c(s)(1+k)$$

or at least

$$\sum_{k\geq 2} a_k(s)\|\varphi_k\|_2^2 \gtrsim_s \|\varphi\|^2_{H^{1/2}}.$$

This would be the torsion analogue of GGRT's negative-definite second variation for their functional $K[F]$ after removing the neutral modes.

5 Entire/Harmonic Approximation Idea

For arbitrary rough domains, u is only useful inside Ω. But after BDV selection, Ω is a smooth near-ball. Then one can try either:

1. Pull back h to the ball and work in the harmonic Dirichlet space there.
2. In dimension two, pull back the holomorphic field

$$F_\Omega = \partial_z u + \bar{z}/4$$

   to the disk and work in a Bergman/Hardy space.
3. Approximate the pulled-back harmonic or holomorphic object by global harmonic polynomials or entire functions, using Runge-type approximation, while tracking boundary norms.

The third option is attractive but risky: approximation alone may not preserve positivity of the level sets or the torsion boundary condition. The safer version is to use approximation only after the selected minimizer has already entered the graph regime, so the geometry is controlled independently.

6 Relation to the Hybrid Penalized Route

This analytic route should be viewed as a possible replacement for the last compactness-heavy step, not for the selection principle.

The proposed chain is:

1. Use the quantitative BDV selection lemma to replace Ω by a selected minimizer U, preserving Q_α.
2. Use density and flatness to put ∂U into a global graph regime.
3. Convert u_U to the harmonic object
$$h_U = u_U + \frac{|x|^2}{2N}$$
or, in $N = 2$, to the holomorphic object
$$F_U = \partial_z u_U + \bar{z}/4.$$
4. Compute a GGRT-style second variation for $\mathcal{K}_s$ or for the level-set deficit in this native space.
5. Use the profile-gap and superlevel replacement ideas to transfer the result to the original asymmetry.

If successful, this would give a more constructive proof of the final near-ball contradiction and might also expose why the exponent 2 is sharp.

7 Most Concrete First Calculation

The first calculation to attempt is:

> For nearly spherical Ω_ε, compute the second variation of
> $$\mathcal{K}_s(\Omega) = \int_{\{u_\Omega > u_\Omega^*(s)\}} u_\Omega \, dx$$
> at the ball, for one fixed $s \in (0, |B|)$.

Do it in spherical harmonics. The expected neutral modes are:

- $k = 0$, removed by volume normalization;
- $k = 1$, removed by translation/barycenter normalization.

If the coefficients for $k \geq 2$ have a positive lower bound comparable to the $H^{1/2}$ weights, then the GGRT variational strategy has a genuine torsion analogue.